

Endterm & Final Exam: SRE Capstone Project - Production Readiness Review

Maximum Score: 100 points (Scales to both Endterm and Final Exam weights in LMS)

Form of Submission: Group project report (PDF), GitHub Repository, and Live Presentation/Defense

Objective

To integrate all theoretical knowledge and practical skills gained throughout the course. Students will design, deploy, and present a production-ready service infrastructure that incorporates Infrastructure as Code (IaC), CI/CD, observability, and automation practices.

Team Roles (Groups of 3–4)

You will act as an SRE team preparing a newly developed microservice or e-commerce platform for a Production Readiness Review (PRR).

Requirements & Architecture

Step 1: Infrastructure as Code (IaC) & Configuration

- Use Terraform to provision infrastructure
- Infrastructure must be reproducible from scratch
- Properly manage Terraform variables and state

Deliverable: terraform directory containing .tf files

Step 2: Continuous Integration & Deployment (CI/CD)

- Configure CI/CD using GitHub Actions, GitLab CI, or Jenkins
- Automatically build Docker images
- Push images to a registry
- Automatically deploy updates to the cluster

Deliverable: Pipeline configuration file and screenshot of successful execution

Step 3: Observability & Alerting

- Deploy Prometheus and Grafana
- Configure metrics scraping
- Create Grafana dashboards visualizing SLIs

- Configure Alertmanager rules

Deliverable: Dashboard screenshots and alert screenshots

Step 4: SRE Operations (SLOs, Scaling & Testing)

- Define SLIs and SLOs
- Configure auto-scaling policies
- Perform load testing using Locust or Apache Benchmark

Deliverable: Screenshots showing scaling during traffic spikes

Submission Instructions

GitHub Repository: Public repository with all project code and configurations.

Written Report (PDF): Technical report with architecture, SLOs, scaling strategy, and screenshots.

Live Presentation & Defense: Demonstrate the system live and answer technical questions.

Detailed Grading Rubric (100 Points Total)

Category	Specific Criteria	Points
Infrastructure as Code (IaC)	Providers, reproducibility, and state management.	15
CI/CD Pipeline	Automated build, registry integration, and deployment.	15
Observability	Metrics collection, Grafana dashboards, and alerting.	15
SRE Operations	SLOs, auto-scaling, and load testing.	15
Documentation	Well-structured report and clean GitHub repository.	10
Defense & Demo	Live demo, technical depth, and individual contribution.	30